Java Coding Standards Example

Recommended by Oracle and commonly followed in the software industry.

The program intentionally includes examples of proper **naming conventions**, **indentation**, **comments**, **constants**, **class structure**, **methods**, **variables**, and **coding style**.

```
/**
 * Program Name : StudentManagementSystem
 * Description  : Demonstrates Java Coding Standards and Naming Conventions.
 *
 * Author       : Your Name
 * Date         : 29-July-2026
 * Version      : 1.0
 */

public class StudentManagementSystem {

    // Constant (UPPER_CASE)
    private static final String UNIVERSITY_NAME = "GLA University";

    // Instance Variables (camelCase)
    private int studentId;
    private String studentName;
    private double marks;

    /**
     * Parameterized Constructor
     *
     * @param studentId Unique Student ID
     * @param studentName Name of the Student
     * @param marks Student Marks
     */
    public StudentManagementSystem(int studentId, String studentName, double
marks) {
        this.studentId = studentId;
        this.studentName = studentName;
        this.marks = marks;
    }

    /**
     * Displays student information.
     */
    public void displayStudentDetails() {
        System.out.println("University : " + UNIVERSITY_NAME);
        System.out.println("Student ID : " + studentId);
        System.out.println("Student Name : " + studentName);
        System.out.println("Marks : " + marks);
        System.out.println("Grade : " + calculateGrade());
    }

    /**
     * Calculates the student's grade.
     *
     * @return Grade based on marks
```

```
     */
    public String calculateGrade() {

        if (marks >= 90) {
            return "A+";
        } else if (marks >= 80) {
            return "A";
        } else if (marks >= 70) {
            return "B";
        } else if (marks >= 60) {
            return "C";
        } else {
            return "Fail";
        }
    }

    /**
     * Main Method
     */
    public static void main(String[] args) {

        // Local Variable
        StudentManagementSystem student =
                new StudentManagementSystem(101, "Rahul Sharma", 88.5);

        student.displayStudentDetails();
    }
}
```

Java Coding Standards Demonstrated

1. Class Naming

- Use **PascalCase (Upper Camel Case)**.
- Class names should be nouns.

```
StudentManagementSystem
Employee
BankAccount
```

---

2. Method Naming

- Use **camelCase**.
- Method names should begin with a verb.

```
displayStudentDetails()
calculateGrade()
printReport()
```

---

3. Variable Naming

- Use **camelCase**.

```
studentId
studentName
totalMarks
averageSalary
```

---

4. Constant Naming

- Use **UPPER_CASE_WITH_UNDERSCORES**.

```
MAX_SIZE
PI
UNIVERSITY_NAME
DEFAULT_TIMEOUT
```

---

5. Package Naming

- Always lowercase.

```
com.gla.frontend
edu.gla.student
org.example.project
```

---

6. Indentation

Use **4 spaces** per indentation level.

```
if (marks >= 40) {
    System.out.println("Pass");
}
```

---

7. Braces

Opening brace should be on the same line.

```
public void display() {
    System.out.println("Hello");
}
```

8. Meaningful Variable Names

✔ Good

```
double annualSalary;
int employeeAge;
```

 Bad

```
double x;
int a;
```

9. Comments

Single Line

```
// Calculate total marks
```

Multi-Line

```
/*
 * This method calculates
 * the final grade.
 */
```

Documentation (Javadoc)

```
/**
 * Returns student's grade.
 *
 * @return Grade
 */
```

10. Proper Spacing

✔ Good

```
int sum = number1 + number2;
```

 Bad

```
int sum=number1+number2;
```

11. Access Modifiers

Declare variables as `private` and expose them through methods if needed.

```
private int studentId;
private String studentName;
```

---

12. Avoid Magic Numbers

✔ Good

```
private static final int PASS_MARKS = 40;
```

□ Bad

```
if (marks >= 40)
```

---

13. Method Length

Keep methods **short** and focused on a single task.

```
calculateGrade()

displayStudentDetails()

calculatePercentage()
```

---

14. Class Organization

A typical Java class should be organized as follows:

```
1. Package Declaration
2. Import Statements
3. Class Declaration
4. Constants
5. Instance Variables
6. Constructors
7. Public Methods
8. Private Helper Methods
9. Main Method (if applicable)
```

Summary of Java Naming Conventions

Element	Convention	Example
Package	lowercase	`com.gla.student`
Class	PascalCase	`StudentManagementSystem`
Interface	PascalCase	`Printable`
Method	camelCase	`calculateGrade()`
Variable	camelCase	`studentName`
Constant	UPPER_CASE	`MAX_MARKS`
Enum	PascalCase	`Department`
Enum Constants	UPPER_CASE	`ACTIVE`, `INACTIVE`

This example is suitable for introducing students to professional Java coding practices and aligns with the coding standards typically followed in industry and recommended by Oracle.